Surface Metrics Comparison: Human-AI vs Human-Human vs AI-AI

Cross-Condition Text Descriptives on the Exchange-Aligned Structure

PENPAL Project

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Contents

1 Overview

This notebook compares surface-level text metrics across conditions using the long-format, exchange-aligned metric tables.

The underlying processed data are now one row per **speaker contribution**, with recovered role metadata:

- **Human / AI** in Human-AI
- **Starter-side / Non-starter-side** in Human-Human and AI-AI

Those role labels are useful for **within-condition description**, but they are not the primary cross-condition comparison axis. For cross-condition comparison, this notebook uses **Slot 1 / Slot 2 asymmetry**.

The notebook therefore no longer assumes old wide `author_1_*` / `author_2_*` columns.

Primary cross-condition object. Per story and per metric m,

$$A_m(i) \;=\; \left|\,\overline{m}_{\text{slot}=1,\,i} - \overline{m}_{\text{slot}=2,\,i}\,\right|.$$

This is the unsigned, label-invariant side-difference magnitude. For each metric we test $A_m \sim$ condition with ANOVA + Kruskal and apply BH adjustment across the metric panel. The LMM slot-marginal contrasts are retained as supplemental. Starter-mechanism and Human-AI speaker-type decompositions are deferred.

Article-use note. The primary result is the per-story unsigned side-difference test. Slot-marginal LMMs and role plots are useful for explaining where the difference comes from, but they should not replace the label-invariant cross-condition result.

2 Load Data

Table 1: Loaded surface-metric rows by condition

condition	n_stories	n_rows
Human-AI	100	1502
Human-Human	36	576
AI-AI	40	640

3 Metric Selection

Table 2: Surface metrics selected for the comparison notebook

metric_alias	column_name
word_count	n_tokens
sentence_length	sentence_length_mean
lexical_diversity	proportion_unique_tokens
readability	flesch_reading_ease
coherence	first_order_coherence

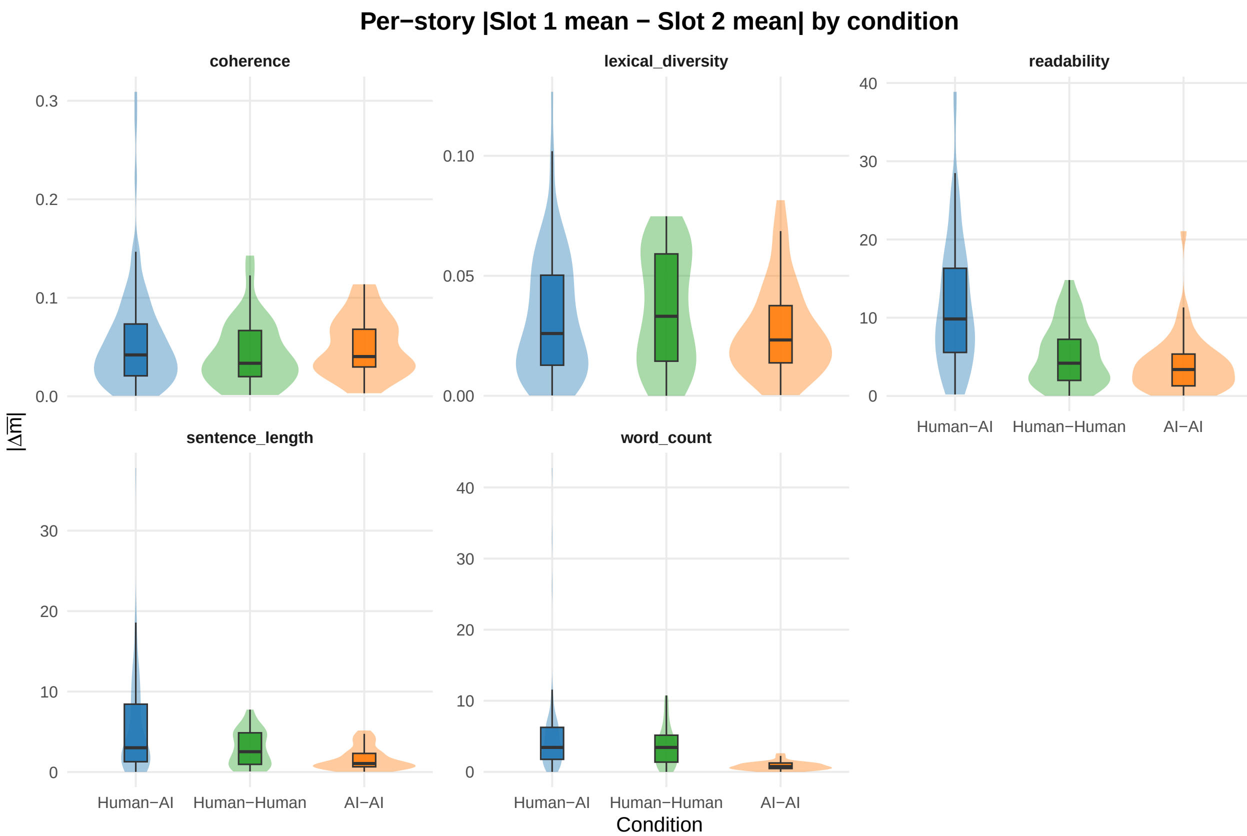
4 Primary: Per-Story Side-Mean Difference

Table 3: Per-story \|side-mean difference\| by condition, per metric

metric	condition	n	mean_A	median_A	sd_A
coherence	Human-AI	95	0.056	0.042	0.053
coherence	Human-Human	36	0.043	0.034	0.033
coherence	AI-AI	37	0.051	0.040	0.030
lexical_diversity	Human-AI	100	0.032	0.026	0.026
lexical_diversity	Human-Human	36	0.036	0.033	0.024
lexical_diversity	AI-AI	40	0.027	0.023	0.019
readability	Human-AI	100	11.751	9.838	8.441
readability	Human-Human	36	5.216	4.184	3.881
readability	AI-AI	40	4.098	3.378	3.851
sentence_length	Human-AI	100	5.371	3.022	5.885
sentence_length	Human-Human	36	2.930	2.531	2.188
sentence_length	AI-AI	40	1.682	1.062	1.406
word_count	Human-AI	100	4.807	3.438	5.934
word_count	Human-Human	36	3.580	3.438	2.786
word_count	AI-AI	40	0.831	0.750	0.583

Table 4: Primary cross-condition tests for A_m ~ condition, per metric (BH-adjusted across metrics)

metric	anova_p	kw_p	anova_p_BH	kw_p_BH
coherence	0.3256	0.4208	0.3256	0.4208
lexical_diversity	0.2856	0.3758	0.3256	0.4208
readability	0.0000	0.0000	0.0000	0.0000
sentence_length	0.0001	0.0002	0.0001	0.0004
word_count	0.0001	0.0000	0.0001	0.0000



5 Bootstrap CIs and Effect Sizes per Metric

Table 5: Bootstrap 95% CI on per-condition mean of A per metric (R = 10,000)

metric	condition	n	mean	ci_lower	ci_upper
coherence	Human-AI	95	0.056	0.046	0.067
coherence	Human-Human	36	0.043	0.032	0.054
coherence	AI-AI	37	0.051	0.042	0.061
lexical_diversity	Human-AI	100	0.032	0.027	0.037
lexical_diversity	Human-Human	36	0.036	0.028	0.043
lexical_diversity	AI-AI	40	0.027	0.022	0.033
readability	Human-AI	100	11.751	10.149	13.468
readability	Human-Human	36	5.216	4.017	6.524

metric	condition	n	mean	ci_lower	ci_upper
readability	AI-AI	40	4.098	3.026	5.384
sentence_length	Human-AI	100	5.371	4.262	6.596
sentence_length	Human-Human	36	2.930	2.232	3.641
sentence_length	AI-AI	40	1.682	1.270	2.126
word_count	Human-AI	100	4.807	3.768	6.076
word_count	Human-Human	36	3.580	2.722	4.510
word_count	AI-AI	40	0.831	0.656	1.009

Table 6: Pairwise Cliff's delta on A per metric

metric	contrast	n_a	n_b	cliff_delta	magnitude
coherence	Human-AI vs Human-Human	95	36	0.125	negligible
coherence	Human-AI vs AI-AI	95	37	-0.032	negligible
coherence	Human-Human vs AI-AI	36	37	-0.176	small
lexical_diversity	Human-AI vs Human-Human	100	36	-0.110	negligible
lexical_diversity	Human-AI vs AI-AI	100	40	0.073	negligible
lexical_diversity	Human-Human vs AI-AI	36	40	0.188	small
readability	Human-AI vs Human-Human	100	36	0.506	large
readability	Human-AI vs AI-AI	100	40	0.625	large
readability	Human-Human vs AI-AI	36	40	0.189	small
sentence_length	Human-AI vs Human-Human	100	36	0.187	small
sentence_length	Human-AI vs AI-AI	100	40	0.424	medium
sentence_length	Human-Human vs AI-AI	36	40	0.312	small
word_count	Human-AI vs Human-Human	100	36	0.092	negligible
word_count	Human-AI vs AI-AI	100	40	0.745	large
word_count	Human-Human vs AI-AI	36	40	0.700	large

6 Planned Contrast: HA vs (HH + AA)/2 per Metric

Table 7: Planned HA vs (HH+AA)/2 contrast per metric, BH-adjusted across metrics

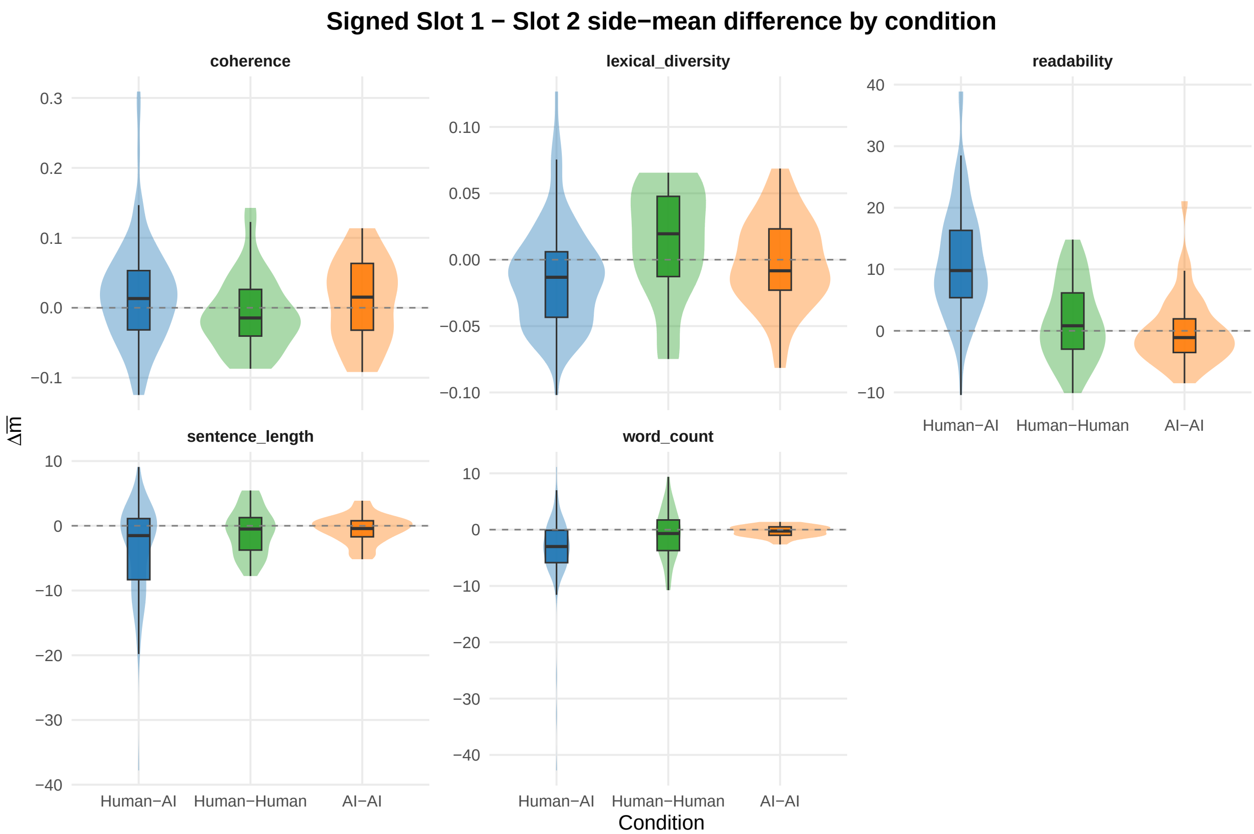
metric	contrast	estimate	se	t	df	p	ci_lower	ci_upper	p_BH
coherence	Human-AI vs (Human-Human + AI-AI)/2	0.009	0.007	1.384	156.313	0.168	-0.004	0.022	0.211
lexical_diversity	Human-AI vs (Human-Human + AI-AI)/2	0.001	0.004	0.232	161.012	0.817	-0.006	0.008	0.817
readability	Human-AI vs (Human-Human + AI-AI)/2	7.094	0.954	7.437	146.219	0.000	5.209	8.979	0.000
sentence_length	Human-AI vs (Human-Human + AI-AI)/2	3.065	0.626	4.896	123.176	0.000	1.826	4.304	0.000
word_count	Human-AI vs (Human-Human + AI-AI)/2	2.601	0.639	4.072	124.736	0.000	1.337	3.866	0.000

7 Directional Stability of Signed Side-Mean Differences (per Metric)

Tests whether the *direction* of the slot-1 vs slot-2 side-mean difference is consistent across stories within each condition, per metric. The signed quantity is $\text{signed}_i = \overline{m}_{\text{slot }1,i} - \overline{m}_{\text{slot }2,i}$.

Table 8: Per-metric directional stability of signed side-mean differences

metric	condition	n	mean	sd	stability	sign_consistency	one_sample_t_p	variance_equality_p
coherence	Human-AI	95	0.019	0.075	0.255	0.611	0.015	0.290
coherence	Human-Human	36	-0.008	0.054	0.144	0.694	0.394	0.290
coherence	AI-AI	37	0.010	0.059	0.165	0.595	0.322	0.290
lexical_diversity	Human-AI	100	-0.012	0.039	0.309	0.670	0.003	0.491
lexical_diversity	Human-Human	36	0.013	0.041	0.309	0.639	0.072	0.491
lexical_diversity	AI-AI	40	-0.002	0.033	0.063	0.525	0.692	0.491
readability	Human-AI	100	11.244	9.112	1.234	0.920	0.000	0.006
readability	Human-Human	36	1.397	6.406	0.218	0.500	0.199	0.006
readability	AI-AI	40	-0.046	5.661	0.008	0.600	0.959	0.006
sentence_length	Human-AI	100	-3.550	7.144	0.497	0.610	0.000	0.000
sentence_length	Human-Human	36	-0.920	3.570	0.258	0.583	0.131	0.000
sentence_length	AI-AI	40	-0.482	2.154	0.224	0.550	0.165	0.000
word_count	Human-AI	100	-3.534	6.777	0.521	0.750	0.000	0.001
word_count	Human-Human	36	-0.844	4.496	0.188	0.556	0.268	0.001
word_count	AI-AI	40	-0.269	0.987	0.272	0.525	0.093	0.001

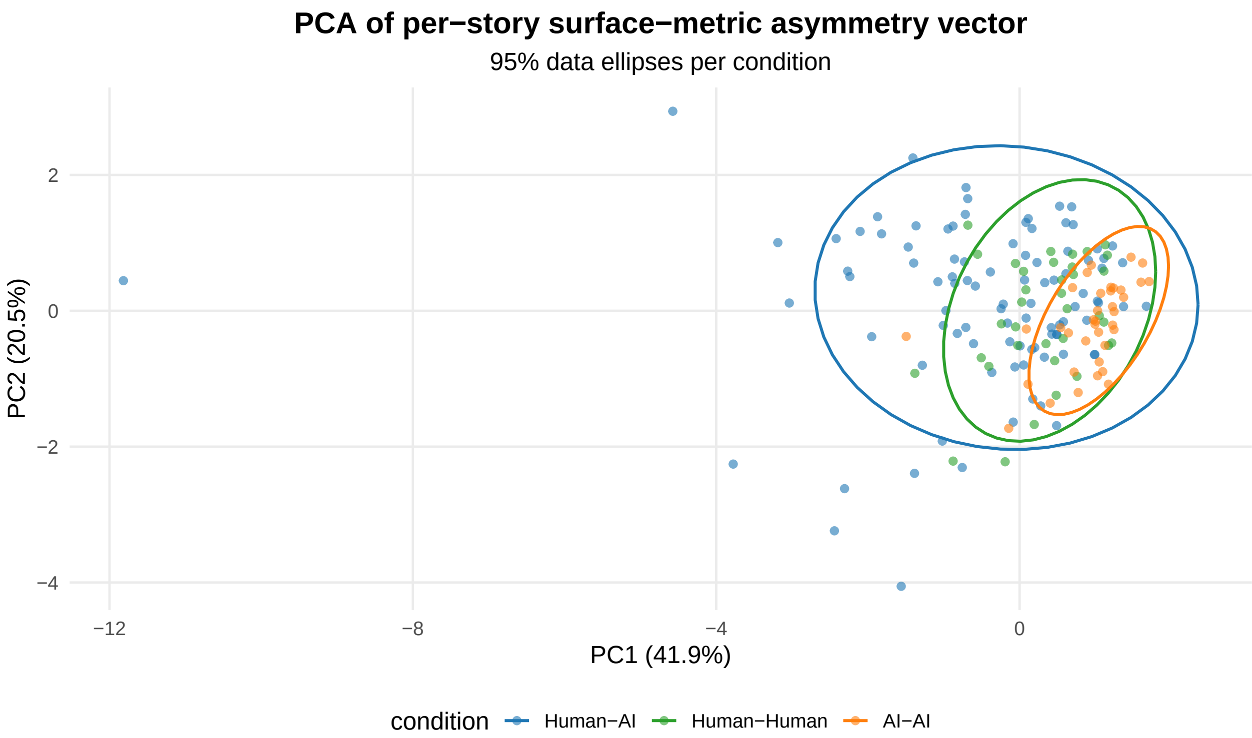


8 Multivariate Composite over Surface Metrics

Stacks the per-story `A_m` vector across all selected surface metrics into one matrix and tests whether the joint condition profiles differ via PERMANOVA (`vegan::adonis2`). A single multivariate answer to "is the cross-metric surface-asymmetry profile distinct in HA?".

```
## Stories with complete A-vector across all 5 metrics: 168
```

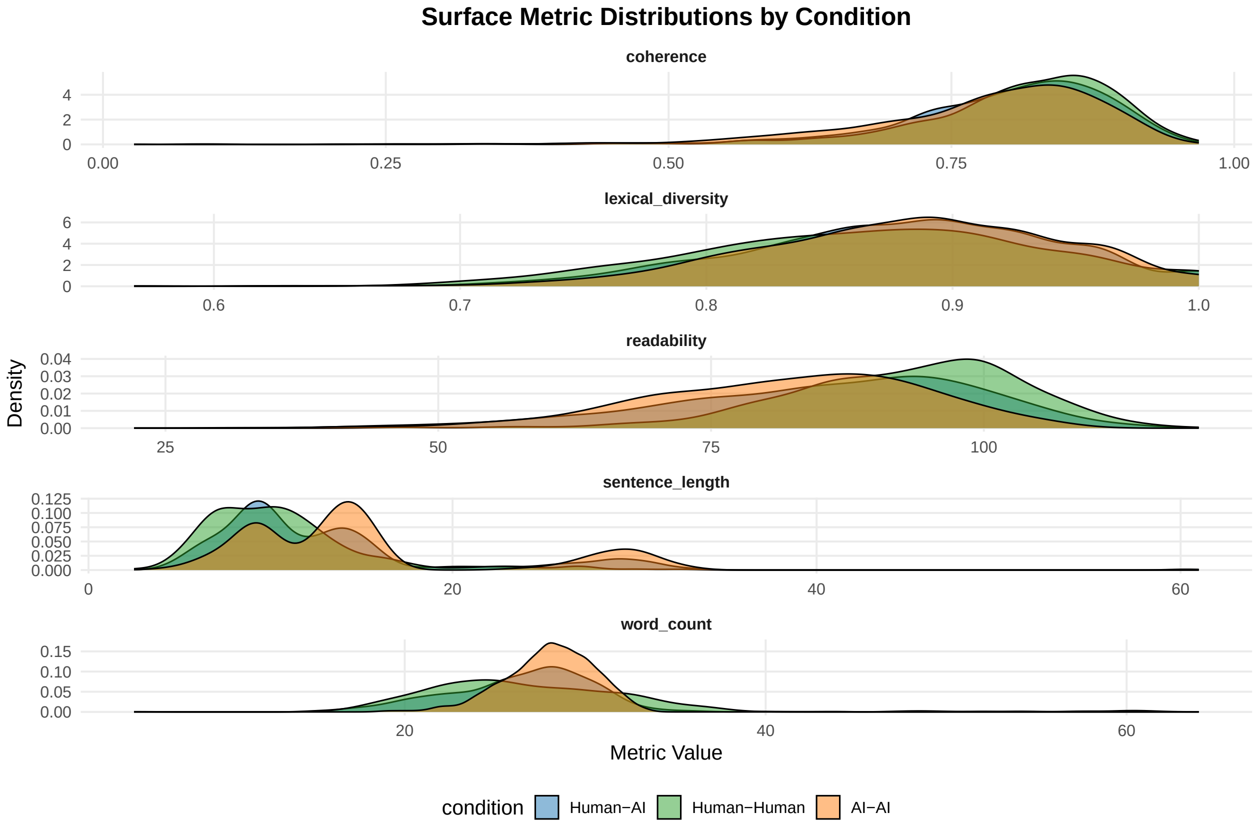
```
## Permutation test for adonis under reduced model
## Permutation: free
## Number of permutations: 9999
##
## vegan::adonis2(formula = X ~ condition, data = A_matrix, permutations = 9999, method = "euclidean")
##           Df SumOfSqs      R2      F Pr(>F)
## Model      2   2608.6 0.15457 15.083  1e-04 ***
## Residual 165  14268.4 0.84543
## Total    167  16877.0 1.00000
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```



9 Supplemental: General Condition-Level Comparison

Table 9: Condition-level descriptives for selected surface metrics

condition	metric	mean_value	sd_value
Human-AI	word_count	27.383	6.303
Human-Human	word_count	26.429	4.831
AI-AI	word_count	28.038	2.439
Human-AI	sentence_length	13.231	7.183
Human-Human	sentence_length	10.719	4.601
AI-AI	sentence_length	15.383	7.398
Human-AI	lexical_diversity	0.879	0.065
Human-Human	lexical_diversity	0.867	0.071
AI-AI	lexical_diversity	0.883	0.062
Human-AI	readability	85.291	14.487
Human-Human	readability	92.794	10.698
AI-AI	readability	81.938	12.515
Human-AI	coherence	0.797	0.100
Human-Human	coherence	0.813	0.086
AI-AI	coherence	0.779	0.108



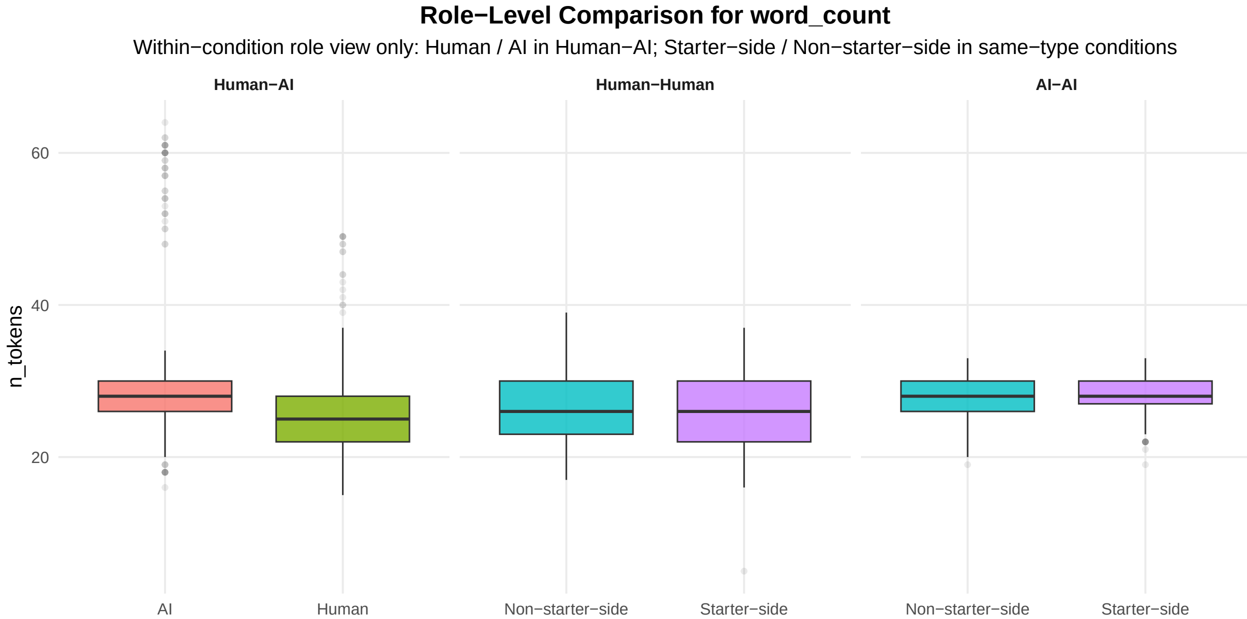

```
##
## === word_count ( n_tokens ) ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: as.formula(paste(col_name, "~ condition + (1 | conversation_id)"))
##    Data: df_combined
##
## REML criterion at convergence: 15843.9
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -5.5437 -0.5168  0.0482  0.5421  5.6212
##
## Random effects:
##  Groups          Name        Variance Std.Dev.
##  conversation_id (Intercept) 12.18    3.49
##  Residual                    16.97    4.12
## Number of obs: 2718, groups:  conversation_id, 176
##
## Fixed effects:
##                       Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)            27.3932     0.3649 172.0176  75.061   <2e-16 ***
## conditionHuman-Human  -0.9644      0.7079 170.6176  -1.362    0.175
## conditionAI-AI         0.6443      0.6814 170.6575   0.946    0.346
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
```

```
##              (Intr) cndH-H
## cndtnHmn-Hm -0.516
## condtnAI-AI -0.536  0.276
##  contrast                   estimate    SE  df t.ratio p.value
##  (Human-AI) - (Human-Human)    0.964 0.708 172   1.362  0.5246
##  (Human-AI) - (AI-AI)         -0.644 0.681 172  -0.946  1.0000
##  (Human-Human) - (AI-AI)      -1.609 0.836 172  -1.924  0.1680
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: bonferroni method for 3 tests
##
## === sentence_length ( sentence_length_mean ) ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: as.formula(paste(col_name, "~ condition + (1 | conversation_id)"))
##    Data: df_combined
##
## REML criterion at convergence: 17348.9
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.9371 -0.5841 -0.1379  0.3873  5.9702
##
## Random effects:
##  Groups          Name        Variance Std.Dev.
##  conversation_id (Intercept) 15.89    3.986
##  Residual                    30.08    5.484
## Number of obs: 2718, groups:  conversation_id, 176
##
## Fixed effects:
##                      Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)           13.1872     0.4230 174.6343  31.172  < 2e-16 ***
## conditionHuman-Human  -2.4683     0.8200 172.7709  -3.010  0.00300 **
## conditionAI-AI         2.1958     0.7894 172.8240   2.782  0.00601 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) cndH-H
## cndtnHmn-Hm -0.516
## condtnAI-AI -0.536  0.276
##  contrast                   estimate    SE  df t.ratio p.value
##  (Human-AI) - (Human-Human)     2.47 0.820 172   3.010  0.0090
##  (Human-AI) - (AI-AI)          -2.20 0.789 172  -2.782  0.0180
##  (Human-Human) - (AI-AI)       -4.66 0.968 172  -4.817  <.0001
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: bonferroni method for 3 tests
##
## === lexical_diversity ( proportion_unique_tokens ) ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: as.formula(paste(col_name, "~ condition + (1 | conversation_id)"))
##    Data: df_combined
```

```
##
## REML criterion at convergence: -7251.7
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.6628 -0.6301  0.0532  0.6793  2.5621
##
## Random effects:
##  Groups          Name        Variance  Std.Dev.
##  conversation_id (Intercept) 0.0005537 0.02353
##  Residual                    0.0037269 0.06105
## Number of obs: 2718, groups:  conversation_id, 176
##
## Fixed effects:
##                         Estimate Std. Error         df t value Pr(>|t|)
## (Intercept)             0.878901   0.002833 175.673984 310.229   <2e-16 ***
## conditionHuman-Human  -0.011641   0.005466 170.670732  -2.130   0.0346 *
## conditionAI-AI         0.003626   0.005262 170.812794   0.689   0.4917
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) cndH-H
## cndtnHmn-Hm -0.518
## condtnAI-AI -0.538  0.279
##  contrast                    estimate      SE  df t.ratio p.value
##  (Human-AI) - (Human-Human)  0.01164 0.00547 171   2.130  0.1039
##  (Human-AI) - (AI-AI)       -0.00363 0.00526 171  -0.689  1.0000
##  (Human-Human) - (AI-AI)    -0.01527 0.00644 169  -2.369  0.0568
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: bonferroni method for 3 tests
##
## === readability ( flesch_reading_ease ) ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: as.formula(paste(col_name, "~ condition + (1 | conversation_id)"))
##    Data: df_combined
##
## REML criterion at convergence: 21403.5
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.4598 -0.6003  0.0989  0.6762  2.7954
##
## Random effects:
##  Groups          Name        Variance Std.Dev.
##  conversation_id (Intercept)  39.36    6.274
##  Residual                    138.51   11.769
## Number of obs: 2718, groups:  conversation_id, 176
##
## Fixed effects:
##                       Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)            85.3663     0.6973 174.6106 122.428  < 2e-16 ***
```

```
## conditionHuman-Human   7.4274     1.3491 171.4946   5.505 1.32e-07 ***
## conditionAI-AI        -3.4287     1.2987 171.5833  -2.640  0.00905 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) cndH-H
## cndtnHmn-Hm -0.517
## condtnAI-AI -0.537  0.277
##  contrast                   estimate   SE  df t.ratio p.value
##  (Human-AI) - (Human-Human)    -7.43 1.35 172  -5.505  <.0001
##  (Human-AI) - (AI-AI)           3.43 1.30 172   2.640  0.0272
##  (Human-Human) - (AI-AI)       10.86 1.59 171   6.819  <.0001
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: bonferroni method for 3 tests
##
## === coherence ( first_order_coherence ) ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: as.formula(paste(col_name, "~ condition + (1 | conversation_id)"))
##    Data: df_combined
##
## REML criterion at convergence: -4215.7
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -7.6178 -0.4534  0.1675  0.6468  2.5294
##
## Random effects:
##  Groups          Name        Variance  Std.Dev.
##  conversation_id (Intercept) 0.0009308 0.03051
##  Residual                    0.0088357 0.09400
## Number of obs: 2321, groups:  conversation_id, 176
##
## Fixed effects:
##                        Estimate Std. Error         df t value Pr(>|t|)
## (Intercept)            0.796751   0.004067 173.275650 195.900   <2e-16 ***
## conditionHuman-Human   0.015752   0.007667 156.790859   2.054   0.0416 *
## conditionAI-AI        -0.019766   0.007713 167.499390  -2.563   0.0113 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) cndH-H
## cndtnHmn-Hm -0.530
## condtnAI-AI -0.527  0.280
##  contrast                   estimate      SE  df t.ratio p.value
##  (Human-AI) - (Human-Human)  -0.0158 0.00767 159  -2.054  0.1247
##  (Human-AI) - (AI-AI)         0.0198 0.00771 169   2.562  0.0338
##  (Human-Human) - (AI-AI)      0.0355 0.00923 160   3.848  0.0005
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: bonferroni method for 3 tests
```

10 Supplemental: Role-Level Comparison




```
##
## === Role contrast for word_count (HH and AA only) ===
##
## Human-Human
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## as.formula(paste(col_name, "~ condition_role + (1 | conversation_id)"))
##    Data: df_subset
##
## REML criterion at convergence: 3350.7
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.3310 -0.7214 -0.0547  0.6854  2.7336
##
## Random effects:
##  Groups          Name        Variance Std.Dev.
##  conversation_id (Intercept)  5.822   2.413
##  Residual                    17.596   4.195
## Number of obs: 576, groups:  conversation_id, 36
##
## Fixed effects:
##                              Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)                   26.7535     0.4720  46.9291  56.678   <2e-16 ***
## condition_roleStarter-side  -0.6493     0.3496 539.0000  -1.857   0.0638 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr)
## cndtn_rlSt- -0.370
```

```
##  contrast                             estimate   SE  df t.ratio p.value
##  (Non-starter-side) - (Starter-side)    0.649 0.35 539   1.857  0.0638
##
## Degrees-of-freedom method: kenward-roger
##
## AI-AI
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## as.formula(paste(col_name, "~ condition_role + (1 | conversation_id)"))
##    Data: df_subset
##
## REML criterion at convergence: 2787.3
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.4842 -0.5970  0.0307  0.6407  3.1584
##
## Random effects:
##  Groups          Name        Variance Std.Dev.
##  conversation_id (Intercept) 2.040    1.428
##  Residual                    3.963    1.991
## Number of obs: 640, groups:  conversation_id, 40
##
## Fixed effects:
##                             Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)                  28.0312     0.2518  47.8650 111.342   <2e-16 ***
## condition_roleStarter-side   0.0125     0.1574 599.0000   0.079    0.937
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr)
## cndtn_rlSt- -0.313
##  contrast                             estimate    SE  df t.ratio p.value
##  (Non-starter-side) - (Starter-side)  -0.0125 0.157 599  -0.079  0.9367
##
## Degrees-of-freedom method: kenward-roger
##
## === Role contrast for sentence_length (HH and AA only) ===
##
## Human-Human
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## as.formula(paste(col_name, "~ condition_role + (1 | conversation_id)"))
##    Data: df_subset
##
## REML criterion at convergence: 3387.9
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -1.7534 -0.6473 -0.1865  0.3213  5.0050
##
```

```
## Random effects:
##  Groups          Name        Variance Std.Dev.
##  conversation_id (Intercept)  0.8361  0.9144
##  Residual                    20.3664  4.5129
## Number of obs: 576, groups:  conversation_id, 36
##
## Fixed effects:
##                            Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)                 10.8688     0.3065  87.9179  35.461   <2e-16 ***
## condition_roleStarter-side  -0.2997     0.3761 539.0000  -0.797    0.426
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr)
## cndtn_rlSt- -0.613
##  contrast                            estimate    SE  df t.ratio p.value
##  (Non-starter-side) - (Starter-side)      0.3 0.376 539   0.797  0.4258
##
## Degrees-of-freedom method: kenward-roger
##
## AI-AI
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## as.formula(paste(col_name, "~ condition_role + (1 | conversation_id)"))
##    Data: df_subset
##
## REML criterion at convergence: 3963.3
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.6516 -0.6469  0.0127  0.4888  3.6673
##
## Random effects:
##  Groups          Name        Variance Std.Dev.
##  conversation_id (Intercept) 31.79    5.638
##  Residual                    23.71    4.870
## Number of obs: 640, groups:  conversation_id, 40
##
## Fixed effects:
##                            Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)                 15.4914     0.9321  42.5466  16.621   <2e-16 ***
## condition_roleStarter-side  -0.2168     0.3850 599.0000  -0.563    0.574
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr)
## cndtn_rlSt- -0.207
##  contrast                            estimate    SE  df t.ratio p.value
##  (Non-starter-side) - (Starter-side)    0.217 0.385 599   0.563  0.5736
##
## Degrees-of-freedom method: kenward-roger
```

```
##
## === Role contrast for lexical_diversity (HH and AA only) ===
##
## Human-Human
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## as.formula(paste(col_name, "~ condition_role + (1 | conversation_id)"))
##    Data: df_subset
##
## REML criterion at convergence: -1410.7
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.2241 -0.6634  0.0779  0.7130  2.2261
##
## Random effects:
##  Groups          Name        Variance  Std.Dev.
##  conversation_id (Intercept) 0.0004353 0.02086
##  Residual                    0.0046488 0.06818
## Number of obs: 576, groups:  conversation_id, 36
##
## Fixed effects:
##                             Estimate Std. Error        df t value Pr(>|t|)
## (Intercept)                8.661e-01  5.314e-03 6.792e+01 163.003   <2e-16 ***
## condition_roleStarter-side 2.254e-03  5.682e-03 5.390e+02   0.397    0.692
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr)
## cndtn_rlSt- -0.535
##  contrast                            estimate      SE  df t.ratio p.value
##  (Non-starter-side) - (Starter-side) -0.00225 0.00568 539  -0.397  0.6918
##
## Degrees-of-freedom method: kenward-roger
##
## AI-AI
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## as.formula(paste(col_name, "~ condition_role + (1 | conversation_id)"))
##    Data: df_subset
##
## REML criterion at convergence: -1819.2
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.0155 -0.6330  0.0427  0.6424  2.2021
##
## Random effects:
##  Groups          Name        Variance  Std.Dev.
##  conversation_id (Intercept) 0.0008464 0.02909
##  Residual                    0.0029921 0.05470
```

```
## Number of obs: 640, groups:  conversation_id, 40
##
## Fixed effects:
##                              Estimate Std. Error        df t value Pr(>|t|)
## (Intercept)                 8.818e-01  5.524e-03 5.428e+01 159.644   <2e-16 ***
## condition_roleStarter-side 1.467e-03  4.324e-03 5.990e+02   0.339    0.735
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr)
## cndtn_rlSt- -0.391
##  contrast                            estimate      SE  df t.ratio p.value
##  (Non-starter-side) - (Starter-side) -0.00147 0.00432 599  -0.339  0.7346
##
## Degrees-of-freedom method: kenward-roger
```

11 Summary

Primary cross-condition result: per-metric $A_m(i) = |\overline{m}_{\text{slot }1,i} - \overline{m}_{\text{slot }2,i}|$, tested with ANOVA + Kruskal and BH-adjusted across metrics.

Robustness layers added:

- bootstrap 95% CIs on per-condition mean of A_m, per metric
- pairwise Cliff's δ effect sizes per metric
- planned single-df contrast HA vs (HH+AA)/2 per metric, BH-adjusted across metrics
- multivariate composite over the surface-metric panel via PERMANOVA + PCA biplot

Supplemental views retained:

- condition-level distributions and marginal `condition` LMMs
- within-condition role comparisons restricted to HH / AA (Starter-side / Non-starter-side); HA Human-vs-AI role LMM lives in analysis/human-ai/surface_metrics_analysis.Rmd and is not duplicated here

Deferred to a follow-up pass:

- starter-mechanism analysis (signed, starter-normalized side differences across all conditions)

Analysis completed: 2026-05-01 23:18:57.418851